

MELBOURNE REGIONAL OFFICE, AustraliaWMO: **948680**Lat: **37.8075S**Lon: **144.9700E**Elev: **106**StdP: **14.64**Time Zone: **10.00 (AUS)**Period: **94-14**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	40.4	42.3	29.7	23.9	64.4	32.0	26.6	58.7	19.8	54.7	17.4	54.0	2.3	90	0.373

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	16.7	95.0	65.8	90.4	64.9	85.9	64.2	70.2	83.2	68.4	81.8	67.0	79.8	5.8	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	97.2	74.5	64.4	90.7	73.2	62.7	85.4	72.1	34.2	83.6	32.7	81.7	31.6	79.9	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	13.9	11.8	DB	36.4	103.6	1.8	4.2	35.1	106.7	34.1	109.2	33.1	111.5	31.8	114.6	(4)
(5)			WB	35.2	73.3	1.5	2.2	34.1	74.9	33.2	76.1	32.4	77.4	31.3	78.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.7	70.0	70.4	66.8	61.2	56.6	52.8	51.7	53.2	56.4	59.4	63.6	66.4	(6)	
	DBStd	8.34	7.04	6.67	5.97	5.00	4.14	3.48	3.24	3.84	4.79	5.51	6.26	6.46	(7)	
	HDD50	45	0	0	0	0	2	11	18	11	3	0	0	0	(8)	
	HDD65	2233	19	15	45	137	263	365	412	366	263	193	100	55	(9)	
	CDD50	3934	619	572	519	335	205	96	71	110	195	293	409	510	(10)	
	CDD65	649	173	167	100	22	2	0	0	6	20	59	100	100	(11)	
	CDH74	6009	1649	1487	839	148	4	0	0	0	30	222	616	1014	(12)	
	CDH80	2574	810	666	325	28	0	0	0	4	57	222	462	462	(13)	
(14) Wind	WSAvg	4.3	4.5	4.3	4.1	3.9	3.7	4.1	4.2	4.4	4.8	4.5	4.4	4.3	(14)	
(15) Precipitation	PrecAvg	25.0	1.9	1.9	1.6	2.3	2.4	1.8	1.8	2.2	2.2	2.5	2.3	2.3	(15)	
	PrecMax	34.1	6.9	9.4	5.4	7.7	5.3	4.6	3.6	4.3	5.0	5.9	6.5	7.8	(16)	
	PrecMin	13.1	0.1	0.0	0.3	0.4	0.4	0.4	0.4	0.9	0.7	0.6	0.4	0.1	(17)	
	PrecStd	5.2	1.4	2.0	1.0	1.5	1.2	0.9	0.8	0.8	1.1	1.2	1.5	1.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	103.5	99.2	95.2	83.7	73.0	65.6	63.9	70.1	78.1	86.5	92.5	98.4	(19)	
		MCWB	68.0	67.3	65.4	61.1	57.7	55.0	52.4	55.8	57.0	60.1	64.2	64.3	(20)	
	2%	DB	95.2	93.3	87.9	78.0	69.1	62.7	60.9	66.0	72.5	80.0	86.2	91.0	(21)	
		MCWB	65.8	66.5	65.2	59.9	56.7	54.0	51.5	53.4	57.0	58.9	62.6	63.6	(22)	
	5%	DB	89.1	88.4	82.8	74.4	66.4	60.5	59.0	63.1	68.8	75.2	81.0	84.9	(23)	
		MCWB	65.8	65.8	63.7	59.4	55.8	52.6	50.6	51.8	55.2	57.7	61.7	63.2	(24)	
	10%	DB	82.8	83.0	78.0	70.7	64.0	58.7	57.4	60.4	65.7	70.5	76.2	78.8	(25)	
		MCWB	65.4	65.6	62.9	58.5	54.9	51.6	49.7	50.6	53.5	56.3	60.6	62.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.2	72.1	69.4	64.6	60.3	57.8	54.8	56.8	61.3	64.0	68.9	68.9	(27)	
		MCDB	82.1	84.5	83.0	76.8	66.2	62.5	60.7	65.3	71.2	75.8	81.4	82.4	(28)	
	2%	WB	70.6	70.1	67.4	62.8	58.8	55.3	53.1	54.7	58.5	61.5	65.9	66.8	(29)	
		MCDB	84.2	82.4	80.3	71.9	65.4	60.6	58.7	63.9	69.0	73.5	77.9	80.0	(30)	
	5%	WB	68.4	68.6	65.7	61.2	57.3	53.7	51.8	53.0	56.5	59.3	63.9	65.3	(31)	
		MCDB	83.2	80.9	77.5	70.3	64.2	59.0	57.1	60.8	66.7	71.3	76.6	78.2	(32)	
	10%	WB	66.5	67.2	64.2	59.5	56.0	52.4	50.7	51.7	54.6	57.5	61.8	63.6	(33)	
		MCDB	79.9	79.2	75.7	68.5	62.7	57.7	55.9	58.6	63.9	67.9	73.5	76.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	17.3	16.7	15.4	14.6	11.7	10.2	10.2	12.2	14.1	15.7	16.4	17.0	(35)	
		MCDBR	30.0	27.8	25.5	21.5	15.4	12.2	12.1	16.4	19.5	25.2	27.2	29.8	(36)	
	5% WB	MCWBR	8.6	8.1	8.1	8.7	7.6	6.7	6.4	7.8	9.1	9.9	9.6	9.1	(37)	
		MCWBR	22.1	20.5	19.8	17.1	12.4	10.0	10.2	14.5	17.1	20.0	21.0	22.3	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.352	0.357	0.332	0.332	0.314	0.303	0.304	0.316	0.336	0.338	0.349	0.343	(40)	
		taud	2.507	2.501	2.569	2.540	2.576	2.593	2.582	2.537	2.475	2.490	2.487	2.523	(41)	
	Ebn at Noon	Ebn at Noon	311	302	297	277	264	258	265	279	291	305	310	315	(42)	
		Edh at Noon	36	34	30	27	22	20	22	26	32	34	36	35	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2268	1980	1578	1095	723	561	627	887	1267	1695	1996	2236	(44)	
	RadStd	RadStd	142	103	90	64	43	45	36	60	91	136	119	126	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.01	+0.72	N/A	N/A	+0.79	N/A	-25	-275	+349	+126
Regional (40 neighbors)		+0.90	N/A	N/A	+1.53	N/A	N/A	-31	-229	+306	+104

Nomenclature: See separate page